A Few Notes on Maxwell's Equations

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Last update: September 22, 2023

1 Introduction

Maxwell's equations are a set of four fundamental equations that describe the behavior of electric and magnetic fields in classical electromagnetism. They were formulated by the Scottish physicist James Clerk Maxwell in the 19th century and are crucial in understanding the behavior of electromagnetic waves, such as light (an idea previously articulated by Faraday [1]). Maxwell's equations are typically written in differential and integral forms [9].

Maxwell's equations are fundamental to classical electromagnetism and have played a crucial role in the development of modern physics and technology, including the understanding of electromagnetic waves, the design of electrical circuits, and the development of technologies such as radio, television, and wireless communication, to name a few.

The rest of this note is organized as follows: Section 2 provides a few definitions needed to discuss Maxwell's equations. Section 3 discusses the differential form of Maxwell's equations, and Section 4 discusses the integral form of Maxwell's equations. Section **??** discusses a few of the properties of electromagnetic waves. Finally, Section 6 outlines some conclusions and final thoughts.

2 Definitions

Symbol	**What**	**SI Units**	**Notes**
$\mathbf{E}$	Electric field	N/C	Newton/Coulomb
$\mathbf{B}$	Magnetic field	N/m/A	Newton/Meter/Ampere
$\mathbf{J}$	Current density	A/m^2	$Ampere/Meter^2$
ρ	Charge density	C/m^3	$Coulomb/Meter^3$
c	Speed of light in a vacuum	m/s	Meter/Second
ϵ_0	Permittivity of free space	F/m	Farad/Meter
μ_0	Permeability of free space	H/m	Henry/Meter

Table 1: A Few Definitions

2.1 A Bit about Electric and Magnetic Constants

Maxwell's equations describe electric and magnetic fields and their propagation. Three constants are normally used in Maxwell's equations: the speed of light (c), the electric permittivity of free space (ϵ_0), and the magnetic permeability of free space (μ_0).

To calculate the value of these constants, first note that the magnetic permeability of free space is taken to have the exact value $\mu_0 = 4\pi \times 10^{-7} N/A^2$. So μ_0 is given. We also know that the speed of light c is equal to $2.99792458 \times 10^8\ m/s$. So we can consider c as given.

Next, notice ϵ_0 and μ_0 are not independent but rather are related to the speed of light [9]:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \tag{1}$$

Since we know the values of μ_0 and c, we can find the value of ϵ_0 by solving for it in Equation (1):

$$\epsilon_0 = \frac{1}{\mu_0 c^2} \tag{2}$$

Substituting the values of μ_0 and c into Equation (2) we get that $\epsilon_0 = 8.8541878128 \times 10^{-12}\ F/m$.

Finally, note that in the presence of polarizable or magnetic media, the effective constants will have different values. In the case of a polarizable medium, called a dielectric, the comparison is stated as a relative permittivity or a dielectric constant. In the case of magnetic media, the relative permeability may be stated.

Back to definitions. For some field $\mathbf{F}$, we call $\nabla \cdot \mathbf{F}$ the divergence of $\mathbf{F}$, while $\nabla \times \mathbf{F}$ is called the curl of $\mathbf{F}$ [3]. It is worth noting that the standard dot (or inner) product is defined as follows:

Definition 2.1. Standard Inner Product: $\langle \mathbf{a}, \mathbf{b} \rangle$

The inner product (aka dot product or scalar product) of two n-dimensional vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, usually denoted by either $\langle \mathbf{a}, \mathbf{b} \rangle$ or $\mathbf{a} \cdot \mathbf{b}$, is defined to be the scalar value

$$\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{j=1}^{n} a_j b_j$$

Since a_j and b_j are scalars (and scalar multiplication commutes) the inner product commutes:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^{n} a_j b_j = \sum_{j=1}^{n} b_j a_j = \langle \mathbf{b}, \mathbf{a} \rangle \tag{3}$$

This is all good, but here we are using the dot notation for the divergence and the divergence is missing a property that the standard dot product has, namely, commutivity (Equation (3)). Specifically, the standard dot product commutes but the divergence does not:

$$\nabla \cdot \mathbf{F} \neq \mathbf{F} \cdot \nabla \tag{4}$$

One way to see this is to notice that the LHS of Equation (4) is a scalar function while the RHS is a differential operator.

One other set of definitions we will need concerns the unit vectors in $\mathbb{R}^3$. Specifically, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are defined to be the unit vectors in $\mathbb{R}^3$ in the x, y, and z directions respectively [2]. Note that $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the canonical basis vectors for $\mathbb{R}^3$ and have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \ \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \ \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3 Maxwell's Equations (differential form)

The differential form of Maxwell's equations assumes the absence of magnetic or polarizable media since these media can change the values of the permittivity and the permeability (see Section (2.1)). The differential form of Maxwell's equations is:

1. Gauss' law for electricity [7]

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

A stationary charge (ρ) creates an electric field ($\mathbf{E}$)

2. Gauss' law for magnetism [4]

$$\nabla \cdot \mathbf{B} = 0$$

Magnetic monopoles do not exist

3. Faraday's law of induction [5]

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

A time-varying magnetic field ($\mathbf{B}$) creates an electric field ($\mathbf{E}$)

4. Ampere's law (with Maxwell's modification) [8]

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

A flowing electric current ($\mathbf{J}$) creates a magnetic field ($\mathbf{B}$), and
a time-varying electric field ($\mathbf{E}$) creates a magnetic field ($\mathbf{B}$)

4 Maxwell's Equations (integral form)

- Gauss' law for electricity

$$\oint_C \mathbf{E}\, d\mathbf{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

  where Q_{enclosed} is the enclosed charge. This equation implies that the electric field lines do not form a continuous closed path. Said another way: isolated positive or negative charges can exist.

- Gauss' law for magnetism

$$\oint_C \mathbf{B}\, d\mathbf{A} = 0$$

  This equation says that magnetic lines of force form a continuous closed path. In other words, it means that no isolated magnetic monopole can exist.

- Faraday's law of induction

$$\oint_C \mathbf{E}\, d\mathbf{l} = -\frac{d}{dt}\Phi_{\mathbf{B}}$$

  where $\Phi_{\mathbf{B}}$ is the magnetic flux through a closed path. This equation says that the line integral of the electric field around any closed path is equal to the rate of change of magnetic flux through the closed path bounded by the surface.

- Ampere's law (with Maxwell's modification)

$$\oint_C \mathbf{B}\, d\mathbf{l} = \mu_0 I_{\text{enclosed}} + \mu_0\epsilon_0 \frac{d}{dt}\int_S \mathbf{E}\, d\mathbf{A}$$

  where I_{enclosed} is the enclosed current. This equation says that both conduction and displacement current can produce a magnetic field.

5 Electromagnetic Waves

In this section we look briefly at electromagnetic waves and how they propagate, as depicted in Figure 1.

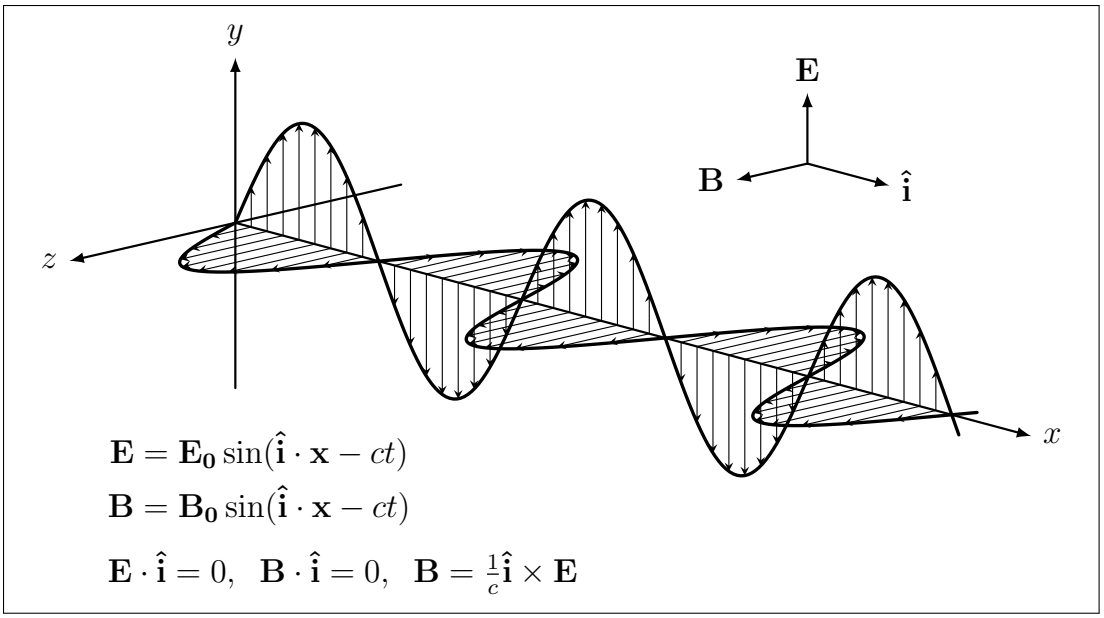

Figure 1: Propagation of an electromagnetic wave in the $\hat{\mathbf{i}}$ direction

6 Conclusions

7 Acknowledgements

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